
Chapter 2:

Factors in Developing Mobile Applications



TOPIC 2: *Factors in Developing Mobile Applications*

- Mobile Software Engineering
- Frameworks and Tools
- Generic UI Development
- Android User.
- More on UIs.
 - VUIs and Mobile Apps i.e. Text-to-Speech Techniques.
 - Designing the Right UI.
 - Multichannel and Multimodal UIs.

Mobile Software Engineering

- In the field of mobile applications development, we primarily develop apps and programs for the following most popular platforms:
 - iOS
 - Android

Mobile Software Engineering

- **Mobile Software Engineering**: is a process of creating software product that is intended to be used by people while they are mobile.
- **Mobile application development**: is the set of processes and procedures involved in writing software for small, wireless computing devices, such as smart-phones and other hand-held devices.
- Factors that influence mobile app development process

1. Distribution Channel

- As a developer, you must first identify which platform best supports your grand idea.
- Although the Google's Play Store and Apple's App Store serve the same function, they have different guidelines→ strictly follow it.
 - ✓ Distribution channel features such as **App. description** and **Image gallery arrangement** play a great role in your app's user engagement.

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2. Audience Research

For your idea to become successful, you must first conduct audience research.

- Knowing your audience involves;
 - Understanding the industry and
 - Getting in-depth knowledge about the people that might use your apps.
- Guess your users' tastes
- Conduct surveys

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3. App. Idea

- Apps don't come out of thin air.
- They start as brilliant ideas.
- Unfortunately, not all ideas become great and successful once you turn them into an app.
- Working on an app development project requires money, effort, and time.
- You must first check if your design is unique and right.
- Most of all, you must be able to differentiate your idea from the what's already available on the platforms.
- It must also be able to resolve the weaknesses of the existing competitive apps.

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4. App. Development Strategy

- To properly execute your app idea, you must come with a plan.
- In this case, you must have a **solid app. development strategy**.
- A cohesive strategy answers all the questions you have towards the app. development process.
- Example
 - Perform deep market research & development process model
 - Use cloud technology
 - Data security mechanism
 - Performance improvement mechanisms
 - Focus on UI/UX
 - Think about the upcoming new technologies

5. UI/UX

- User interface (UI) and user experience (UX) are part of what keeps users hooked on an app.
- Successful apps are those that provide a **hassle-free experience** aside from **serving its purpose**.
- Users prefer apps that are “handier” compared to those that require too much effort to operate.

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6. Testing

- Developing and designing your mobile app idea is only half of the job in your mobile app development journey.
- You must do tests to ensure your design and application runs smoothly.
- It's a way to secure that your mobile app has no exceptions and bugs.
- If you launch your app without testing it, you can run into problems that can spark fires which are difficult to manage.

7. Security

- Security must be part of your app development process.
- Users should not have to think of the threat of malware when they use your app.
- You should leverage the security features of the device's OS.
- You can keep the user's information safe through encryption.
- You should also restrict your app's permissions to a minimum.

Always prioritize security to ensure your app isn't vulnerable to external threats.

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8. Monitoring

- You still have work to do after you publish your mobile app.
- You must be hands-on with fixes, performance improvements, changes, and adding new features.
- As a developer, you must track crashes and look at technical information which helps you to resolve the problems that the users might encounter.

Frameworks and Tools

- In the field of mobile applications development, we primarily develop apps and programs for the following most popular platforms:

- iOS
- Android



Android Development Frameworks

- **React Native**

- Open-source mobile framework that was created by **Facebook** and released in 2015.
- It provides support for various IDEs and mobile app development tools.
- It is used by many companies such as **Instagram, Tesla, Walmart, Baidu** etc. to develop their mobile apps.

- **Flutter**

- Comparatively recent open-source development framework that was released by **Google** in 2017.
- Flutter is Google's portable UI toolkit for building **beautiful, natively-compiled applications for mobile, web, and desktop apps**.
- Written in the **Dart programming language** (Also developed by Google)
- It is currently used by famous companies such as **Alibaba, Tencent, Google**.

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- Ionic
 - Very popular open-source mobile framework that was released in 2013.
 - It's also free
 - Ionic can be used to build and ship beautiful cross-platform hybrid & Progressive Web Apps easily.
 - It is cross-platform framework, which means that you can create apps and then customize them for Android, iOS, Windows, etc.
 - Ionic has documentation support with real-life examples.
- Adobe PhoneGap:
 - Open-source mobile application development framework that was purchased by Adobe Systems in 2011.
 - Its unique feature is that it can be used to build mobile applications using HTML5, CSS3, and JavaScript instead of using APIs for Android, iOS, or Windows Phone.

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- Titanium SDK
 - Open-source framework that was created by [Appcelerator, Inc.](#) and released in 2018.
 - It's quite popular and has been used in around [70, 000+ Mobile Applications](#) till now.
 - The Titanium environment helps organizations get to market 60% faster and achieve a significant competitive advantage.

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- JQuery Mobile

- Touch-Optimized Web Framework that was developed by the [jQuery team](#) and released in 2010.
- It is based on [HTML5](#) and can be used to create responsive web sites and apps that can be accessed using multiple devices such as [smartphones](#), [tablets](#), [desktops](#) etc.
- “write less, do more”
- Provides support for different operating systems such as [Windows](#), [Mac OS](#), [Blackberry](#) etc.

- Corona SDK

- Developed by [Corona Labs Inc.](#) and released in 2009.
- Great platform for creating apps for [mobile and desktop systems](#).
- Corona is [incredibly](#)

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- Native Script

- Open-source framework for creating **native mobile apps** that were developed by **Progress** and initially released in 2014.
- It was developed using **JavaScript/TypeScript**

- Mobile Angular UI

- Mobile Angular UI is a mobile UI framework that allows the creation of **HTML5 hybrid mobile apps** using a combination of **Angular JS and Bootstrap**.

Generic UI Development

A good User Interface (UI) focuses on making user's interactions **simple and efficient**.

- User would appreciate a website with intuitive user interface that leads them towards their task in most engaging way.

UI design focuses on **thinking of a user, what they might need to do when they visit website/app and**

- **ensure that the interface has elements that are easy to access and understand.**

As a UI designer, you need to understand the **goals, skills, preferences & tendencies** of the user to

- make a better interface.

The Generic User Interface (Generic UI, GUI) framework allows you to create UI screens using **Java**

- **and XML.**

XML provides a **declarative** approach to the screen layout and reduces the amount of code which is

- required for building the user interface.

Android User and History

- Android is the most popular operating system in the world, with over 2.5 billion active users spanning over 190 countries.
- Created by Andy Rubin as the open-source alternative to iPhone.
- Android quickly became the favorite operating system for most mobile manufacturers in the early 2010s.
- Industries such as Samsung, LG, Motorola and HTC are all launching phones running Android.
- It quickly became the most popular mobile OS, hitting billions active users by 2014.
- Android has added more users every year since its inception.

User Interface Design

- User interface (UI) design is the process designers use to build interfaces in software or computerized devices, **focusing on looks or style**.
- UI design refers to graphical user interfaces and other forms, e.g., voice-controlled interfaces.
- User interfaces are the **access points** where users interact with designs.
- They come in three formats:
 - **Graphical user interfaces (GUIs)**: users interact with visual representations on digital control panels. e.g; Desktop UI
 - **Voice-controlled interfaces (VUIs)**: users interact with these through their voices.
 - Smart assistants, e.g., Siri on iPhone and Alexa on Amazon devices are VUIs.
 - **Gesture-based interfaces**: users engage with 3D design spaces through bodily motions: e.g., in virtual reality (VR) games.

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- A user interface can be judged by three main properties:
 - **Ease-of-use**: how easy it is to use a certain user interface
 - **Efficiency for inputting information**: how fast information can be input through the user interface
 - **Efficiency for outputting information**: how fast the user can process the information coming back from the system

VUIs and Mobile Apps, Text-to-Speech Technique

- Voice user interfaces are user interfaces that are used through speech.
 - Typical examples include smart speakers and voice assistants.
- VUIs employ speech recognition & natural language understanding technologies to transform user speech into text and meaning.
- Speech is a tool for enhancing traditional touch user interfaces into multi-modal voice user interfaces.
- Voice user interfaces are highly intuitive as they use the most natural way for us to communicate → through speech.
- They are significantly faster than typing to input information but slower than reading/ seeing to output information from the computer system back to the user.

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- The first voice user interfaces were **IVR**, Interactive Voice Response, systems that enabled users to interact with a phone system by using **speech**.
- Typically, IVRs recognized only digits, but nonetheless they were early voice user interfaces.

Designing the Right UI

- In mobile app development, the **UI and UX** play the most significant roles in a mobile app's success.
- From the perspective of a designer, an exceptional UI for mobile app can only be ensured when you follow some tried and tested UI design principles.
- It is always advisable to stick to certain principles that are accepted by the designers' community.
- Such UIs not only attract the user, but also help in retaining them in the long run.

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- Check the UI principles listed below:
 - **Consistency of the design layout**: different sections of the app must be coherent in the design, and there must be a consistent flow of the layout throughout the app.
 - **Unambiguous interactive elements**: design elements that are interactive in nature must be clearly depicted.
 - **Single trial learning experience**: the term 'Intuitive' is often used to describe mobile apps when suggesting that they are very user-friendly.
 - Users understand the flow of the app without remembering the operational steps.

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- **Anticipate and answer users' queries:** when a user's tap a button, they want to know whether the process has started and how long will it take.
 - Gmail app is a good example of such a design, as it displays messages like “saved in draft, message sent, etc.” to inform the users about the completion of the task.
- **Layered user experience:** UX should be layered.
 - Layered here refers that all the features of an app must not be exposed at once.
 - WhatsApp offers a good layered experience as users explore new features as they go deeper into the app.

Android Features

- It is **open-source**.
- There are **lots of mobile applications** that can be developed by it.
- Android offers a user-friendly interface that can be customized to your liking.
- Provides many interesting features like weather details, opening screen, location services.
- Provides support for messaging services → **SMS & MMS**,
- Web browser → **support many web browsers**
- Storage → **SQLite**,
- Connectivity → **GSM, CDMA, Bluetooth, Wi-Fi**
- Media support → **audio, video, graphics and image**
- Multi-language support → **localization**

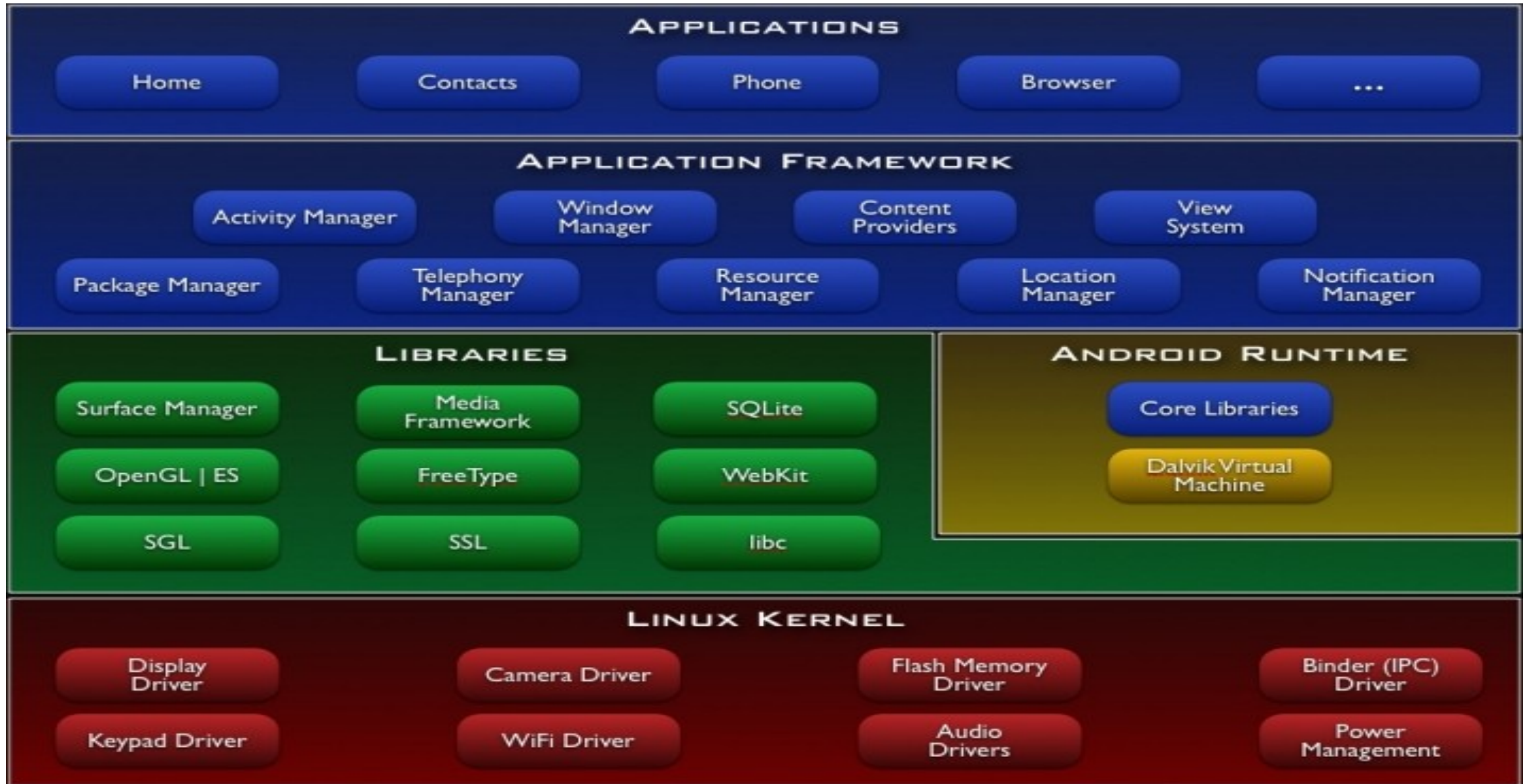
Android Versions

Date	Version	Nickname/Brand	API	Features
Sep 2008	1.0, 1.1	Android	1, 2	
Apr 2009	1.5	Cupcake	3	
Sep 2009	1.6	Donut	4	
Oct 2009	2.0, 2.1	Éclair	5-7	
May 2010	2.2	Froyo	8	
Dec 2010	2.3.x	Gingerbread	9-10	
Feb 2011	3.0.x	Honeycomb	11-13	Tablet only
Oct 2011	4.0.x	Ice Cream Sandwich	14,15	
Jul 2012	4.1-4.3	Jelly Bean	16-18	
Oct 2013	4.4, 4.4W	KitKat	19, 20	Kitkat for Wearables

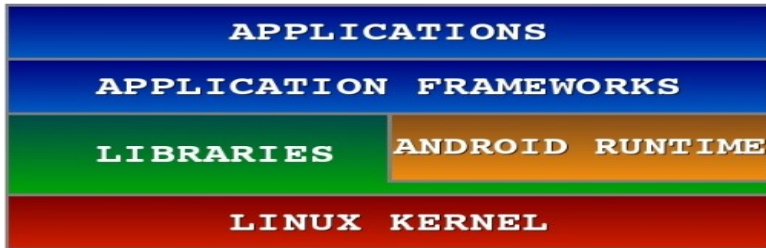
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Date	Version	Nickname/Brand	API	Features
Nov 2014	5.0, 5.1	Lollipop (L)	21, 22	Material Design, ART runtime - Just In Time vs Ahead Of Time
Oct 2015	6.0	Marshmallow (M)	23	Runtime permissions, Doze mode
Aug 2016	7.0, 7.1	Nougat (N)	24, 25	Multi window, Doze on the go, Vulkan API, Quick settings tiles, App shortcut
Aug 2017	8.0, 8.1	Oreo (O)	26, 27	Picture in picture, Custom & Downloadable fonts, Neural Network API, Programmatic safe browsing actions, Video thumbnail extractor, Wallpapers color API, Fingerprint updates, Cryptography update
Aug 2018	9.0	Pie (P)	28	Display cutout, Notifications – messaging, Multi camera support, ImageDecoder, Animation(GIF, WebP animated images), HDR VP9, HEIF and Media APIs
Sep 2019	10	Q (Android Q)	29	Gesture nav, Smart replies, Foldable devices, More privacy control
Sep 2020	11	Red Velvet Cake (R)	30	Chat bubbles, Notification history, One time permission, Permission auto-reset, 5G detection API
Oct 2021	12, 12L	Snow Cone (S)	31, 32	Material You - design refresh, Splash screen API, Bluetooth permissions (Scanning nearby devices doesn't require location permission), Optimize for large screens & foldables
Aug 2022	13	Tiramisu (T)	33	Notification permission, Nearby Wi-Fi devices permission

Android Architecture



Android S/W Stack - Application



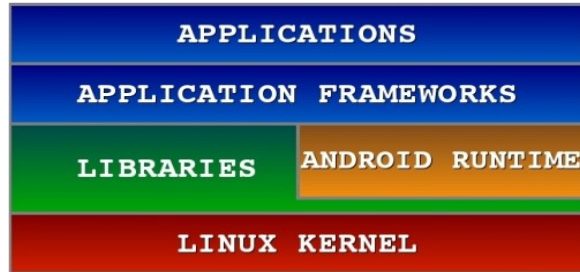
- Android provides a set of core applications:
 - Email Client
 - SMS Program
 - Calendar
 - Maps
 - Browser
 - Contacts
 - etc...
- All applications are written using the Java language.

Applications (Written in Java code)

- *Android Play Store*
- *Entertainment*
- *Productivity*
- *Personalization*
- *Education*
- *Geo-communication*
-

Android S/W Stack - Application

Enabling and simplifying the reuse of components



- Developers have full access to the same framework APIs (Application programming interface: set of routines, protocols and tools for building software application) used by the core applications.
- Users are allowed to replace components.

Feature	Role
View System	Used to build an application, including lists, grids, text boxes, buttons, and embedded web browser
Content Provider	Enabling applications to access data from other applications or to share their own data
Resource Manager	Providing access to non-code resources (localized strings, graphics, and layout files)
Notification Manager	Enabling all applications to display customer alerts in the status bar
Activity Manager	Managing the lifecycle of applications and providing a common navigation back stack

Android S/W Stack - Application



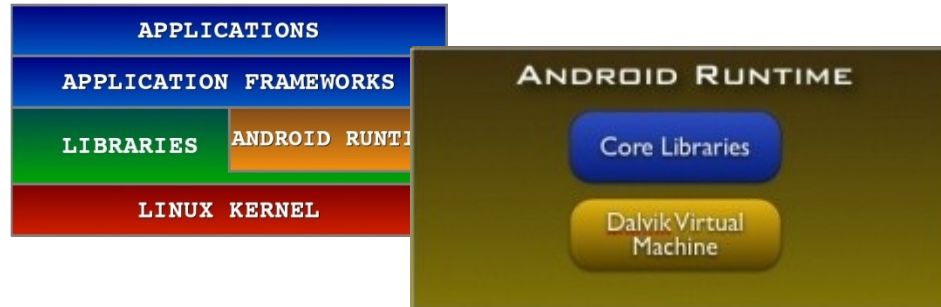
Native Libraries

(C/C++ code)

- **Graphics** (Surface Manager)
- **Multimedia** (Media Framework)
- **Database DBMS** (SQLite)
- **Font Management** (FreeType)
- **WebKit**
- **C libraries** (Bionic)
-

- Including a set of C/C++ libraries used by components of the Android system
- Exposed to developers through the Android application framework.

Android S/W Stack - Application



- **Core Libraries**

- ✓ Providing most of the functionality available in the core libraries of the Java language
- ✓ APIs
 - Data Structures
 - Utilities
 - File Access
 - Network Access
 - Graphics
 - Etc

Dalvik Virtual Machine (VM)

- **Novel** Java Virtual Machine implementation (not using the Oracle JVM)
- **Open License** (Oracle JVM is not open!)
- **Optimized** for memory-constrained devices
- **Faster** than Oracle JVM
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Android S/W Stack - Application

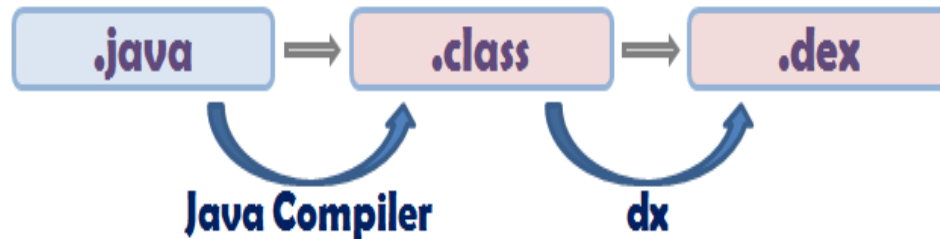
- **Dalvik Virtual Machine**

- ✓ Providing environment on which every Android application runs
 - Each Android application runs in its own process, with its own instance of the Dalvik VM.
 - Dalvik has been written such that a device can run multiple VMs efficiently.

- ✓ Register-based virtual machine

- ✓ Executing the Dalvik Executable (.dex) format

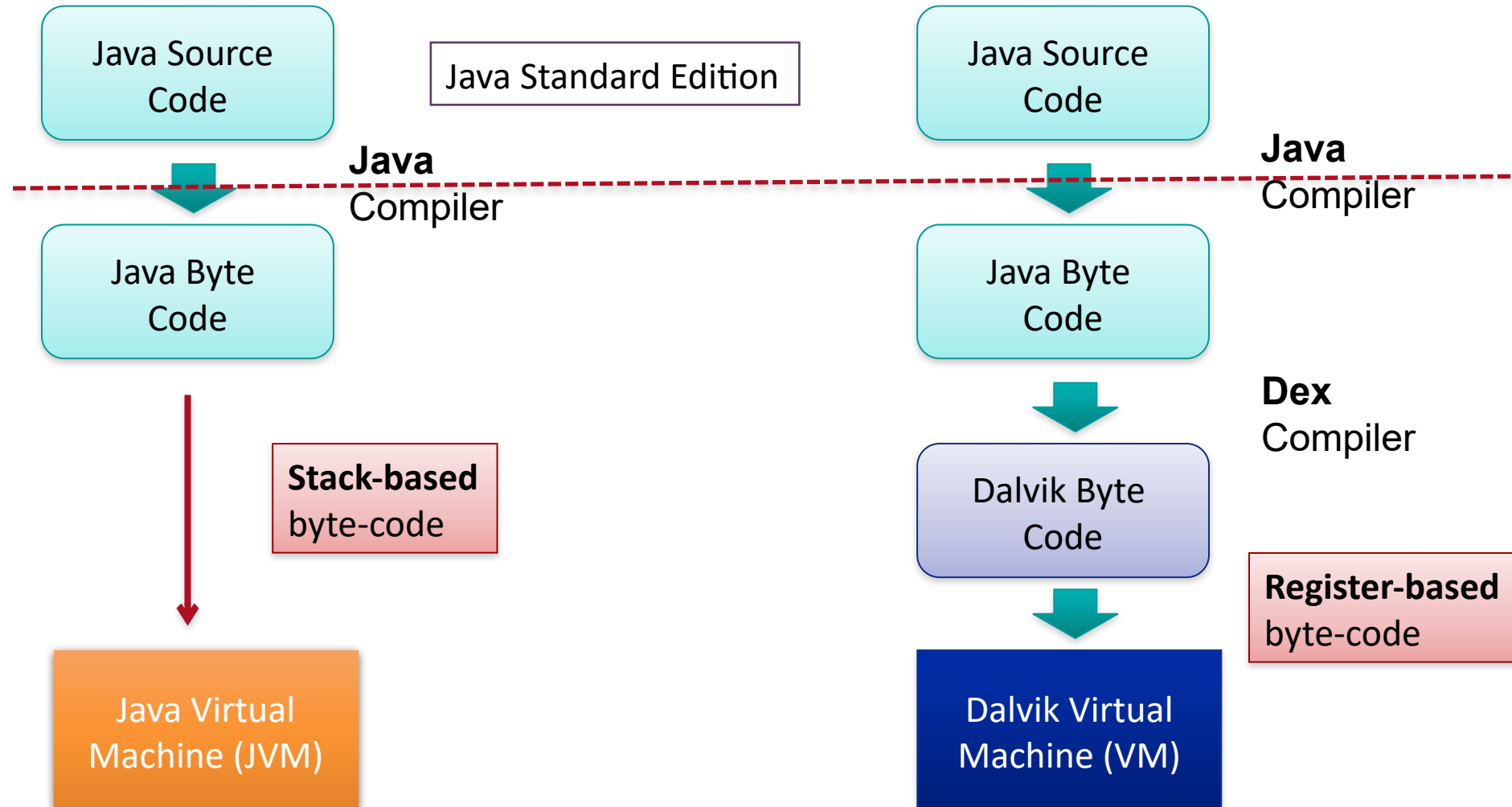
- .dex format is optimized for minimal memory footprint.
 - Compilation



- ✓ Relying on the Linux Kernel for:

- **Threading:** enables the DVM to effectively manage concurrent execution with android application
 - **Low-level memory management:** The Linux kernel handles low-level memory management tasks, including memory allocation and deallocation

Android S/W Stack - Application



Android S/W Stack - Application



- Relying on Linux Kernel 2.6 for core system services
 - Memory and Process Management
 - Network Stack
 - Driver Model
 - Security
- Providing an abstraction layer between the H/W and the rest of the S/W stack

Built on top of **Linux kernel** (v. 2.6-3.0)

Advantages:

- **Portability** (i.e. easy to compile on different hardware architectures)
- **Security** (e.g. secure multi-process environment)
- **Power Management**

Thank You!



Questions